

Robust Machine Text Detection under Domain Shift via MMD-Guided Semi-Supervised Learning

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1 Introduction

With the widespread deployment of large-scale language models, distinguishing between human-written and machine-generated text has become a critical task in NLP. This capability supports content authenticity, curbs misuse, and enhances transparency in human–AI interaction. However, detection remains challenging due to variations in text across domains and ongoing distributional shifts driven by rapid adoption of generative tools.

In this work, we address machine-generated text detection under such realistic conditions. We propose two complementary deep learning approaches: a domain-specific pipeline that separates domain and label prediction, and a unified, domain-agnostic model designed for robust, cross-domain detection. Experiments on multi-domain datasets demonstrate the effectiveness and generalizability of our framework.

2 Methodology

Dataset We consider a supervised binary classification task to distinguish human-written text from machine-generated content. The training data spans two domains: Domain 1 is balanced (500 human vs. 500 machine), while Domain 2 is heavily imbalanced (4,750 machine vs. 250 human) across 5,000 samples. During testing, domain labels are unavailable, and predictions must generalize across both domains. This setting presents two key challenges: domain shift due to stylistic variation, and severe class imbalance, particularly in Domain 2, motivating the two complementary modeling strategies introduced in this research.

Domain-specific Modeling To address domain heterogeneity, we first adopt a domain-specific modeling framework that decomposes the task into two stages: domain prediction followed by domain-specific classification. This serves as a strong and well-motivated baseline, grounded in the clear inter-domain differences such that Domain 1 is relatively small and balanced, whereas Domain 2 is significantly larger and heavily skewed toward machine-generated content. This partitioned setup enables each classifier to specialize in domain-specific characteristics. A notable strength of this approach is its ability to model semantic divergence, wherein the same label may correspond to different linguistic realizations across domains. This reflects a conditional distribution shift in $P(Y|X)$, shaped by domain-specific factors. Independent training allows the model to learn domain-conditioned label semantics, resulting in more consistent and interpretable predictions. Furthermore, the framework simplifies optimization by eliminating the need for global alignment and enabling localized resampling where appropriate. Its primary limitation lies in its reliance on accurate domain prediction at inference, which may propagate errors if domain boundaries are ambiguous. Nevertheless, this method offers a principled and effective solution in scenarios characterized by domain divergence and label imbalance.

Domain-unified Modeling In real-world scenarios, textual data often spans multiple domains with implicit boundaries and shifting distributions, making domain-specific modeling costly and hard to scale. This motivates a unified framework capable of generalizing across domains without explicit partitioning. To this end, we propose a domain-agnostic model that integrates domain adaptation into representation learning, enabling robust prediction under domain shift and label imbalance. At the core is a representation alignment mechanism based on Maximum Mean Discrepancy (MMD) [1], which promotes domain-invariant but label-discriminative features. Given two sets of embedded representations $\mathcal{X} \sim \mathcal{P}$ and $\mathcal{Y} \sim \mathcal{Q}$, the squared MMD is defined as: $\text{MMD}^2(\mathcal{X}, \mathcal{Y}) = \left\| \frac{1}{n} \sum_{i=1}^n \phi(x_i) - \frac{1}{m} \sum_{j=1}^m \phi(y_j) \right\|_{\mathcal{H}}^2$, where $\phi(\cdot)$ maps into a reproducing kernel Hilbert space $\mathcal{H}$. In practice, we apply MMD to the embedding layer using a Gaussian kernel. Specifically, MMD is applied between same-label embeddings across domains, such as $\text{MMD}^2(D_1^{\text{human}}, D_2^{\text{human}}) + \text{MMD}^2(D_1^{\text{AI}}, D_2^{\text{AI}})$. It encourages the model to produce similar embeddings for human-written or machine-generated texts regardless of their domain origin. This class-conditional alignment reduces intra-class variation caused by domain shift, helping the model form more compact and consistent clusters for each label. At the same time, by pulling same-label samples closer across domains, the representation space naturally separates different classes more clearly, improving inter-class discriminability. This process parallels supervised contrastive learning, but instead of aligning samples within the same domain, we focus on aligning semantically equivalent instances across domains. To ensure that alignment occurs over task-relevant features, the MMD loss is applied to deeper representations where class semantics are more explicitly encoded. Moreover, we delay its activation using a warm-up schedule, allowing the encoder to first stabilize under cross-entropy supervision. This prevents early-stage disruption and allows MMD to function as a refinement mechanism that sharpens class boundaries while suppressing domain-induced noise. In addition, we incorporate a lightweight domain- and label-aware resampling

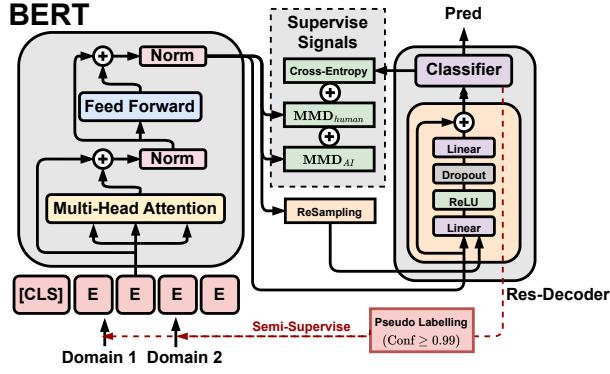


Figure 1: Overview of our BERT-based MMD unified model architecture.

strategy to mitigate class imbalance during batch construction. To further leverage unlabeled data, we incorporate a semi-supervised learning component tailored to our task. Since AI- and human-generated texts often differ in distributional density and surface variability, labeled data alone may not fully capture the boundary between them. By generating pseudo-labels for high-confidence unlabeled samples and enforcing consistency under perturbations, the model learns to stabilize its predictions near natural clusters in representation space. This helps reveal decision boundaries aligned with the underlying data geometry. In summary, each training step jointly optimizes classification loss, MMD alignment, and semi-supervised consistency, enabling the model to generalize effectively across domains and label distributions without relying on explicit domain supervision at inference time.

Model Architecture We adopt an encoder-only architecture based on a randomly initialized BERT [2], as illustrated in Figure 1. This design is motivated by the need to detect subtle differences in writing style, fluency, and structure between human- and machine-generated text. BERT’s bidirectional attention enables it to capture unnatural phrasing, repetition, and abrupt transitions, which are typical of AI-generated content. Even without pretraining, the architecture proves effective under supervision. Each tokenized input text is preprocessed and resampled before being passed to the BERT encoder. We extract the [CLS] token from the final hidden layer as a sentence-level representation. This embedding is then passed through a two-layer feedforward network with a residual connection, which serves to refine the representation for the downstream task. While BERT captures contextual information, its output may not align directly with task-specific decision boundaries. The feedforward block helps re-shape the representation to emphasize relevant discriminative features, while the residual path retains core contextual signals. A final linear layer produces the prediction logits. Together, these components functionally serve as a shallow decoder that maps general-purpose embeddings to task-specific decision space. This architecture is shared across all experiments. In the unified framework, it is trained end-to-end for cross-domain prediction. In the domain-specific setting, the same encoder is used, but separate classifiers are trained for each domain. This consistent backbone facilitates comparison across frameworks while fully leveraging BERT’s capacity for stylistic and structural representation learning.

3 Experiments

Experiment Setups All models and experimental pipelines are implemented using PyTorch. Experiments are conducted on a 16-core AMD EPYC 9654 CPU and a single NVIDIA RTX 4090 GPU.

Implementation To enhance model performance, we apply several training and preprocessing strategies. Input texts are processed with TF-IDF weighting and n-gram tokenization, then padded to a fixed length of 256. To address class imbalance, we experiment with random duplication, SMOTE, and Mixup. Early stopping is used based on validation performance. For semi-supervised learning, high-confidence (≥ 0.99) validation predictions are added as pseudo-labeled data, and the model is then fine-tuned with the encoder frozen.

Evaluation We evaluate six models, including a MLP baseline, a single BERT classifier, two domain-specific frameworks (using either MLP or LSTM classifiers), and our unified model with and without semi-supervised learning. All models are assessed using accuracy and macro-averaged F1 score on the held-out validation set to reflect both overall correctness and balance across classes.

Main results Table 1 summarizes the overall performance of all models. Among the baselines, the BERT_{base} model achieves strong results, even without explicit processing on domain discrepancy. This demonstrates that even without external knowledge, the architecture itself is effective at capturing linguistic patterns and style-related cues that are useful for cross-domain generalization. In contrast, the shallow MLP_{base} baseline underperforms, reflecting the limited representational capacity of simpler models when applied to heterogeneous input distributions. The domain-specific models further improve performance by explicitly partitioning the task according to domain. While the LSTM variant performs slightly worse than the MLP classifier, both approaches benefit from reduced intra-domain variance and highlight the utility of modeling domain-conditioned label semantics. This suggests that although the approach is

Table 1: Overall experiment results.

Model	Acc	F-score
MLP _{base}	0.8423	0.7756
BERT _{base}	0.9267	0.9060
D-Specific _{MLP}	0.9350	0.8259
D-Specific _{LSTM}	0.8900	0.8059
D-Unified _{supervised}	0.9352	0.9122
D-Unified_{semi-supervised}	0.9817	0.9583

Table 2: Ablation study on our final model.

Model Variant	Acc	F-score
D-Unified_{semi-supervised}	0.9817	0.9583
w/o MMD Loss	0.9336	0.9295
Resampling	Acc	F-score
SMOTE (ori)	0.9817	0.9583
Mixup	0.9773	0.9482

architecturally simple, domain-aware decomposition offers a practical and effective solution. The unified model, trained without domain supervision, performs comparably in the fully supervised setting, but shows reduced generalization. This is likely due to overfitting to the dominant domain and insufficient exposure to minority patterns. Semi-supervised learning helps mitigate this by leveraging high-confidence pseudo-labeled samples from the validation set, which enrich the training signal in underrepresented regions. This correction to the learned decision boundary enables the model to more effectively bridge domain and distributional gaps, ultimately yielding the best performance overall.

Ablation Study To assess the contribution of each component within our unified framework, we conduct a set of ablation experiments. As shown in Table 1, removing the semi-supervised module leads to a substantial decline in both accuracy and F1 score. This highlights a broader insight in representation learning: when the training data distribution is skewed or incomplete, models trained purely under supervised signals may fail to sufficiently capture minority or low-density regions of the input space. In contrast, semi-supervised learning, by incorporating high-confidence pseudo-labeled samples, can implicitly recover underrepresented patterns and guide the model to refine its decision boundary. These pseudo-labeled instances are likely to populate low-density or minority areas of the input space, providing additional learning signals that reduce decision boundary bias toward dominant classes or domains. In our case, pseudo-labels from validation data help bridge the gap between training and inference distributions. From table 2 We also observe that removing the MMD loss significantly impairs performance, confirming its importance in promoting cross-domain alignment. By encouraging representations of same-label instances from different domains to converge, the model is guided toward learning domain-invariant but label-discriminative features, validating the essential property for robust generalization across heterogeneous sources. Finally, we investigate the role of resampling techniques. Our default choice, SMOTE, improves model performance by generating synthetic minority samples through local linear interpolation in feature space. Substituting SMOTE [3] with Mixup [4], another strategy that forms convex combinations of randomly selected input-label pairs, leads to a slight performance drop. This suggests that while resampling is beneficial overall, the exact strategy plays a secondary role. This is likely because the semi-supervised framework already addresses distributional imbalance by augmenting the training signal in structurally meaningful ways.

4 Conclusion

We present a comparative study on machine-generated text detection under domain shift and data imbalance. Our experimental results demonstrate that while domain-specific modeling offers a simple and effective baseline, a unified architecture equipped with representation alignment and semi-supervised learning achieves superior generalization. The proposed framework requires no domain labels at inference and remains robust across heterogeneous inputs, offering a practical solution for real-world detection scenarios.

References

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